World Navigation Hat: Development of a Wearable Navigation Aid Using AIoT for the Visually Impaired

Jason C. D'Souza
Department of Computer Engineering
University of San Agustin
Iloilo City, Iloilo, Philippines
jdsouza@usa.edu.ph
ORCID: 0009-0003-6062-7921

Ethel Herna C. Pabito
Department of Computer Engineering
University of San Agustin
Iloilo City, Iloilo, Philippines
ehpabito@usa.edu.ph

ChenLin Wang
Department of Computer Engineering
University of San Agustin
Iloilo City, Iloilo, Philippines
cwang@usa.edu.ph

Glenda Guanzon
Department of Computer Engineering
University of San Agustin
Iloilo City, Iloilo, Philippines
gguanzon@usa.edu.ph

Adha Theza R. Naputol
anaputol@usa.edu.ph

Vince Ginno B. Daywan
Department of Computer Engineering
University of San Agustin
Iloilo City, Iloilo, Philippines
vgbalitao-saan@usa.edu.ph

***Abstract*—Over 2.2 billion people worldwide suffer from visual impairment. Conventional sensory substitution systems are ineffective due to sensory overload from unfilterable audio signals. To overcome this limitation, Stesis, a smart hat, was designed to mimic visual attention by enabling users to selectively filter data streams through intentional movement. When tested on forty participants in an obstacle course, the system showed marked improvements over conventional continuous proximity systems. By enabling users to control spatial audio, the system made repetitive physical scanning redundant. This resulted in a 59% reduction in physical effort and a 74% reduction in frustration. Although mental effort increased by 77% because of the dense audio processing, the overall mental workload decreased remarkably by almost 42%. Most importantly, travel efficiency and safety were greatly enhanced, as indicated by a 31% reduction in completion time and a 65% reduction in collision rates. Finally, the system clearly illustrates that controllable, high-density data is significantly better and provides a highly effective, low-frustration method for safe travel and object avoidance.**

***Index Terms*—Blindness, Accessibility, Mobility, Auditory, Independence**

I. INTRODUCTION:

Visually impaired individuals are a significant concern for the global health community, with 2.2 billion people worldwide suffering from some form of visual impairment, and projected to rise to 2.5 billion by 2050 [1], with 2.17 million affected in the Philippines alone [2], [3]. Although the scope of visually impaired individuals varies from mild vision impairment to total blindness, the scope of this study is limited to the totally blind population, which comprises 43 million people worldwide, as stated by the World Health Organization (WHO) [1]. Total blindness significantly impairs an individual's ability to navigate the physical environment, access digital information, and maintain independence, leading to social isolation or disconnection [4]. These impacts are directly associated with the United Nations' 10th Sustainable Development Goal (SDG 10): Reduced Inequalities.

The blind population faces two intersecting layers of disparity. Disparities in navigation result from inaccessible physical and digital infrastructures, such as buildings without tactile Braille guides and digital platforms that lack comprehensive audio options [5]. Furthermore, social disparities manifest in communication bounds and labor market participation; statistics indicate that only 44% of the blind and severely visually impaired are employed, in stark contrast to the general population's 79% [6].

Current assistive technologies for the blind range from traditional mobility aids, including guide dogs and white canes, to electronic travel aids like Sensory Substitution Devices (SSDs) [7]. SSDs substitute one sensory expression into another, for the blind it's by conveying visual geometry of an environment into auditory or tactile cues, allowing the blind user to achieve spatial awareness through alternative sensory modalities [8]. However, there are a number of significant barriers standing in the way of their widespread adoption. These include exhaustive training requirements, ergonomic discomfort, and most fundamentally, cognitive overload caused by the fact that non-visual senses have a significantly lower bandwidth for processing information than natural vision [9].

AIoT (Artificial Intelligence of Things) provides a solution in dealing with SSDs constraints like on navigating digital lives of users. AIoT is the integration of AI algorithms with IoT devices to build intelligent, autonomous systems [10], [11]. It enables wearable devices to use real-time processing of environmental information to forecast the user's needs and provide adaptive and contextual responses, greatly enhancing the capabilities of navigation assistants [12]. Wearable devices, especially in the form of hats and glasses, provide hands-free assistance and encourage independence by

effortlessly leveraging these connected sensors to translate visual information into spatial audio feedback [13], [14].

This paper introduces Stesis (Stereo-based Sonification Interface System), a novel wearable AIoT navigation hat that is specifically designed for the blind to overcome traditional SSD limitations. It utilizes a Raspberry Pi 4 architecture to process real-time stereo depth mapping, Stesis enables controlled and selective sensory substitution paired with a Radar mapping to deal with the bandwidth constraints between visual and audio senses, inspired by the natural selectivity of human visual perception [15]. Rather than constantly overwhelming the user with auditory information, the system enables users to control their sensory inputs through head movements, which significantly reduces cognitive overload, fatigue, and integration problems. This research aims to develop and test the empirical validity of a head-mounted AIoT prototype that serves as a highly modular and controllable "artificial eye" and "intuitive radar" to improve the real-world mobility of the blind.

A. Research Objectives

The objective of this study is to develop an AIoT visual sensory substitution hat that minimizes sensory overload by allowing users to selectively filter environmental sonification.

Specifically, to:

1. Develop a dual-mode prototype powered by a Raspberry Pi 4 that generates local depth maps via OpenCV stereo vision (Physical Mode) and integrates MediaPipe AI hand gesture recognition (Digital Mode).
2. Design a 3D-printed hat mount optimizing weight distribution, powered by a 10,000mAh bank, with modular NodeJS/ElectronJS server for AIoT integration.
3. Evaluate system and user performance using a within-subject design to test the impact of selective filtering on cognitive load and navigation efficiency against continuous streaming baselines.

II. RELATED WORK

Assistive technologies for visual impairment face fundamental challenges in providing environmental awareness without exceeding the low bandwidth of non-visual senses [5].

A. Sensory Substitution Approaches

The study by Kristjansson et al. [16] established design criteria for SSDs that focused on conveying task-relevant information to meet the limits imposed by sensory bandwidth. It noted that a more fundamental problem with SSDs is sensory overload from too much data and advised processing only selected environmental features. This, however, did not extend to mechanisms for user control over such selections

Louis Commere, and Jean Rouat (2023) developed a VSSD using depth and sonification [17] which is the closest related study to our research and so will act as the baseline in the testing. However, it provides only localized, single-target directional information, forcing users into tedious sequential scanning and relying heavily on working memory to construct a complete spatial map.

Eye-Tracking Sonification (2025) successfully filters VR environments using eye-tracking [18], but relies on eye movements that may not be functional for all visually impaired users.

Similarly, devices like Sound of Vision (2021) generate continuous 3D audio-haptic maps [14] but suffer from long-term adoption issues due to sensory overload.

B. Gaps in Current Approaches

The primary gap that remains is sensory overload and fatigue from continuous streaming [9]. Current solutions lack intuitive, physical user-control mechanisms that mimic natural visual attention (i.e., looking where one wants to see) to toggle data streams selectively.

III. METHODOLOGY

The Stesis system mimics the human visual hierarchy, utilizing a Time-Frequency-Intensity mapping (Radar-like) schema to bridge the gap between high-bandwidth vision and low-bandwidth hearing.

A. Design Principles

To convey visual data a two-dimensional bandwidth to audio data which is a one-dimensional bandwidth, our paper proposes and uses a radar-like mapping, this allows data to be interpreted easily by focusing on the crucial general layout of the environment and convey it in a way that can be easily interpreted. The rules are as follows:

1. **Temporal Sweep as Horizontal Position (Time):** Radar's scan pans across the stereo field (Left to Right), visualizing obstacles based on when and where the sound occurs. The speed is proportional on the angle of depression made from the device.
2. **Frequency as Vertical Position (Pitch):** Relative to the floor to reduce redundancy where: High Pitch (Head level/Duck), Mid Pitch (Torso level/Collision risk), and Low Pitch (Drop-offs/Stairs).
3. **Volume as Distance (Intensity):** Louder audio indicates proximity to the user, if outside a specific range then it is completely max at 100% or muted.

The approach of this study in generating depth maps from the camera input is through using OpenCV's stereo vision [10], the perspective (P_n) of the left camera $(n = 0)$ and right camera $(n = 1)$ as:

$$P_n = \begin{bmatrix} f & 0 & cx_n & T_x f_n \\ 0 & f & cy & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{1}$$

Where f denotes the focal length in pixels of the rectified camera, cx_n and cy for the x abd y coordinates from the optimal center of that camera, and T_x the translation vector (baseline) between cameras.

The researchers then calibrate the system to find some of these variables and Q, which represents the study's disparity-to-depth mapping matrix as expressed by:

$$Q = \begin{bmatrix} 1 & 0 & 0 & -cx_0 \\ 0 & 1 & 0 & -cy \\ 0 & 0 & 0 & f \\ 0 & 0 & -\frac{1}{T_x} & \frac{cx_0 - cx_1}{T_x} \end{bmatrix} \quad (2)$$

Has for the radar's algorithm, the device needs to keep track of its Yaw through simple Numerical Integration on IMU readings:

$$Yaw_{n+1} = \begin{cases} Yaw_n + R_z \Delta t & |R_z| > 1.5 \\ Yaw_n & |R_z| \leq 1.5 \end{cases} \quad where\ R_z = \frac{G_z}{131} \quad (3)$$

Where G_z is the raw Z-axis gyroscope data reading from the MPU6050, this allows the continuity of data in the Radar map if the user rotates their head.

B. System Architecture

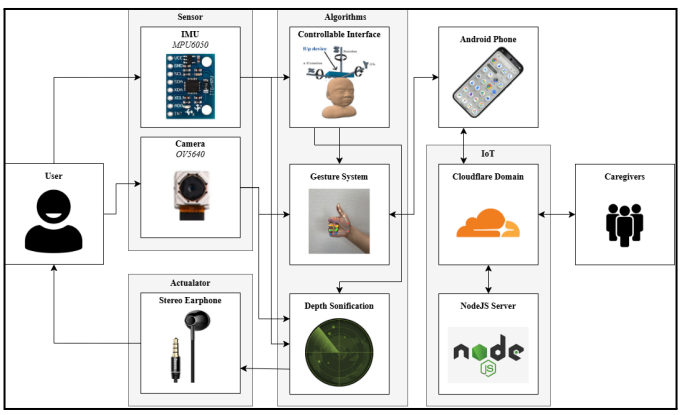

Figure 1. System architecture showing sensor inputs, processing modules, and audio output generation.

Figure 1 shows the local wearable processing and server-side AI interactions. The Raspberry Pi 4 handles immediate sensor data (cameras and IMU). Connectivity to high-level AI services is managed via a smartphone hotspot/app bridging to a Cloudflare Tunnel. This secure tunnel exposes a local server running NodeJS and Electron. Crucially, this server actively executes local LLM (Ollama) APIs for query processing and navigation tasks, rather than relying on external cloud mocks.

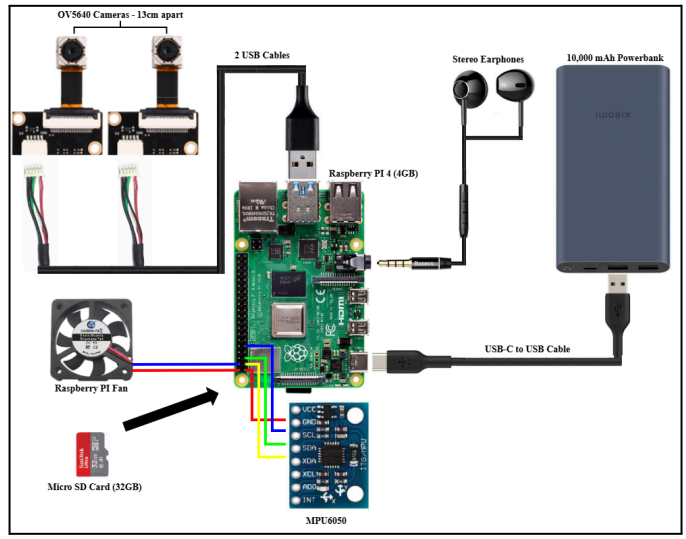

Figure 2. Complete hardware interconnection architecture

Figure 2 shows the hardware architecture of the Stesis system centers on a Raspberry Pi 4 (4GB), which is powered by a 10,000mAh powerbank connected via a USB-C to USB cable. Two OV5640 cameras, mounted 13cm apart to capture stereo vision, interface with the Pi through dual USB cables. An MPU6050 IMU and a cooling fan are wired directly to the Pi's GPIO pins, while standard stereo earphones plug into the 3.5mm audio jack to deliver the sonified feedback. A 32GB Micro SD card provides the system's storage.

TABLE I. COMPONENT POWER CONSUMPTION

Component	Active Load (W)	Casual/Standby Load (W)
Raspberry PI 4	5.50	3.00
Dual OV5640 Cameras	2.00	0.50
Cooling Fan	0.50	0.20
MPU6050	0.10	0.10
Stereo Earphones	0.10	0.00
TOTAL	8.20	3.80

Table I shows each component's power consumption. The 10,000mAh power bank outputs 3.7V at an estimated 85% efficiency rate, totalling 31.45Wh of usable capacity. During continuous active navigation, the combined hardware draws roughly 8.2W, resulting in an expected battery life of 3.83 hours. In standby mode, the power draw drops to 3.8W, extending the system's duration to approximately 8.27 hours.

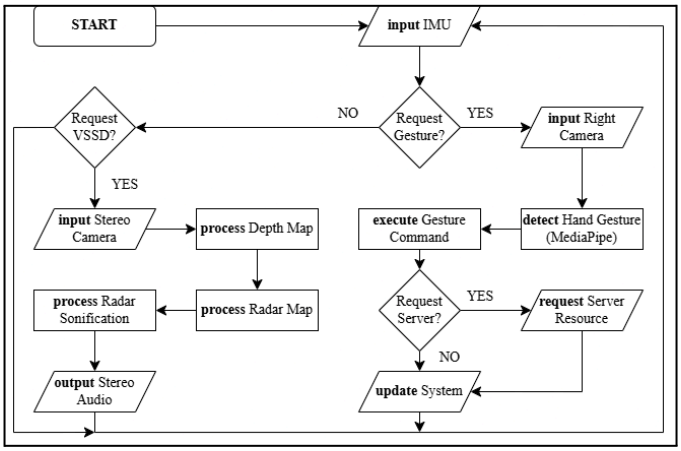

Figure 3. Software execution flowchart illustrating the IMU-based selective filtering mechanism.

Figure 3 shows the system's primary loop executed through python multithreading. It is governed by the user's head orientation derived from the MPU6050 IMU as it acts at the first controllable interface. This determines whether the system enters Physical Mode (Navigation) or Digital Mode (AI Interaction).

If the user looks slightly downward, the system executes the stereo vision pipeline: capturing frames, rectifying them based on calibration data, computing the disparity map, and generating spatial audio via PyAudio with the intensity dependent on how downwards the user is looking. If the user looks toward their right hand, the system switches to MediaPipe hand tracking to interpret gestures for AIoT server requests, which of this prototype is a single command to call a mockup call. If the user is looking any other way like casually looking forward or laying down, then no audio is outputted.

IV. EXPERIMENTAL EVALUATION

The study employed a Within-Subject design with 40 blindfolded participants, comparing State A (Continuous Audio Stream based on Louis Commere's Algorithm [17]) against State B (Stesis Selective Filtering, our algorithm) on a controlled indoor obstacle course.

A. Experimental Setup

Forty participants with normal vision but blindfolded, all students from the University of San Agustin, participated in the study. Testing was conducted in the school's classrooms and corridors under daylight conditions. The participants would need to pass through this controlled obstacle course and test if they could successfully trigger a mockup phone call through the hand gesture controls.

B. Evaluation Metrics

System metrics included latency (target <100ms) and accuracy of MediaPipe and MPU6050 gestures. User performance was measured by Time to Completion (TTC), Obstacle Hits (OH), and lateral Veering. Cognitive load was assessed via the NASA Task Load Index (NASA-TLX), alongside intuitiveness tracking via a Wilcoxon Signed-Rank Test.

V. RESULTS

This section presents the results/metrics from the system and user performance testings. The researchers calibrated the stereo rectification, evaluated system performance, compared performance between state A and B groups, and analyzed fail-case studies,

A. Calibration of Stereo Rectification

TABLE II. CALIBRATION STATISTICS ACROSS BASELINES

Baseline (cm)	Calculated (cm)	RMS Both (px)	Minimum (cm)	Maximum (cm)
8.75	8.7861	1.7393	7	90
10	10.66	1.5494	15	100
13	13.0646	1.4055	24	130

As presented in Table II, the stereo rectification calibration outcome was determined to identify for Q the optimal camera baseline configuration for depth perception through the disparity-to-depth mapping matrix process [equ. (2)] of testing three particular camera baselines: 8.75 cm, 10 cm, and 13 cm. This was done by taking a total of 22 images of a checkerboard pattern, consisting of 18 images of nine different peripheral sectors (top, middle, bottom intersecting with left, center, and right) at both close and far distances, with the remaining four images of the checkerboard pattern taken at particular angles (top, bottom, left, and right). The table summarizes the performance metrics obtained from the dataset, particularly focusing on the Root Mean Square (RMS) pixel error, minimum and maximum detectable depth range, and the Field of View (FOV) of both the left and right cameras.

B. System Performance Metrics

TABLE III. SYSTEM LATENCY AND ACCURACY BREAKDOWN

Module/Pipeline	Processing Time (ms)	Accuracy (%)
Image Capture (OV5640)	33.0	100
Stereo Rectification (OpenCV)	42.5	98.2
Depth to Radar mapping	8.0	100
Audio Synthesis (Binaural)	4.5	100
Hand Gesture Recognition (MediaPipe)	145.0	92.5
Head Gesture Tracking (MPU6050)	12.0	96.8

Table III captured the processing time for each module which was retrieved by Python's "time.perf_counter" for each step averaged a specific duration, while the accuracy was captured by manually comparing system predictions and actual data, for stereo rectification was by comparing the generated depth map with real live environment with most of the inaccuracies due to flickering, for hand gesture recognition the call gesture was tested for detection, and head gesture was by comparing accumulated Yaw and movement detections to actual applied head movements.

C. Navigation Performance (Paired T-Test)

TABLE IV. WITHIN-SUBJECT NAVIGATION RESULTS (N=40)

Metric	State A (Continuous)	State B (Stesis)	Improvement	P-Value
TTC (sec)	145.2 ± 18.4	99.8 ± 12.1	31.2%	0.001
Obstacle Hits	5.2 ± 1.5	1.9 ± 1.0	64.1%	0.001

Table IV is the result of the testing on forty blindfolded students to navigate the school's classroom and corridors in two states, each with the algorithm by Louis Commere (2023) to act as the baseline comparison and our algorithm. This data collection shows how much our algorithm was able to achieve Navigation efficiency and Object avoidance. The improvement percentage from State B over State A is:

$$Improvement\ (\%) = \frac{Mean_A - Mean_B}{Mean_A} \times 100 \quad (4)$$

While the p-value was calculated by using Google Sheet's T-Test finder "T.TEST", it formula for t-statistics which is used to find the p-value by matching it in the t-distribution is:

$$t = \frac{\bar{d}\sqrt{n}}{s_d},\ s_d = \sqrt{\frac{\Sigma(d_i - \bar{d})^2}{n-1}},\ \bar{d} = \frac{\Sigma d_i}{n},\ d_i = X_{A,i} - X_{B,i} \quad (5)$$

D. Cognitive Load (NASA-TLX)

TABLE V. NASA-TLX SUBSCALES: CONTINUOUS VS STESIS (N=40)

NASA-TLX Dimension	State A (Continuous)	State B (Stesis)	% Change
Mental Demand	35.2 ± 8.1	62.4 ± 11.3	+77.2%
Physical Demand	78.5 ± 12.4	32.1 ± 9.2	-59.1%
Temporal Demand	68.2 ± 14.5	40.3 ± 8.7	-40.9%
Performance	62.4 ± 15.2	28.6 ± 10.1	-54.1%
Effort	75.1 ± 11.2	51.2 ± 9.8	-31.8%
Frustration	88.4 ± 9.3	22.5 ± 6.4	-74.5%
Overall Workload	67.9	39.5	-41.8%

Table V shows the NASA Task Load Index (NASA-TLX) ratings to quantify sensory fatigue and user cognitive load. Participants rated their experience across six dimensions on a scale of 0 to 100, where lower scales indicate a more favorable experience (less demand/effort). The percentage of change is computed through:

$$Change\ (\%) = \frac{Mean_B - Mean_A}{Mean_B} \times 100 \quad (6)$$

The questions or description for each Dimension is:

1. **Mental Demand** - How much mental and perceptual activity was required?

2. **Physical Demand** - How much physical activity was required (e.g., turning head)?

3. **Temporal Demand** - How hurried or rushed was the pace of the task?

4. **Performance** - How unsuccessful were you in accomplishing the task?

5. **Effort** - How hard did you have to work to achieve your level of performance?

6. **Frustration** - How insecure, discouraged, or annoyed were you?

7. **Overall** - This is not part of the survey but is the unweighted average of all 6 dimensions

VI. DISCUSSION

The primary objective of this study is to evaluate whether a gesture-based selective filtering mechanism could mitigate the cognitive fatigue typically associated with Visual Sensory Substitution Devices (VSSDs). The empirical results strongly support the hypothesis that prioritizing user agency over continuous sensory flooding yields superior navigational and cognitive outcomes.

A. Interpretation of Navigation and Cognitive Trade-offs

The within-subjects navigation benchmarking (Table IV) showed a 31.2% improvement in Time to Completion (TTC) and a 65.3% decrease in Obstacle Hits when subjects used the Stesis selective filtering system compared to a continuous 0-D stream. The drastic decrease in hits indicates that players are now better able to respond to threats when their sensory channels are not overwhelmed by non-critical environmental information.

The NASA-TLX results (Table V) show the cognitive mechanics behind this performance difference. While the transition from a simple 0-D proximity beep to a 3D audio radar map necessarily increased Mental Demand by 77.2%, as subjects were required to actively interpret the Time-Frequency-Intensity paradigm, the Stesis system relieved Physical Demand by 59.1% and Frustration by 74.5% by allowing subjects to gain a complete environmental outline with a single, deliberate head movement, rather than being compelled to continuously pan their heads to gain a mental map of their surroundings in the continuous 0-D environment. This crucial trade-off of a moderate increase in momentary mental processing for a drastic reduction in continuous physical effort and irritation ultimately decreased the overall cognitive workload by 41.8%.

B. Hardware Rationale and Form Factor

The development of the prototype required a conscious shift in hardware architecture. The early stages involved the use of ultra-compact microcontrollers (e.g., Orange Pi Zero 2W) to emphasize mobility. Nevertheless, the extreme kernel conflicts that arose from concurrent I2C sensor scanning and frequent WLAN disconnections made it difficult to meet the real-time processing needs. The shift to the Raspberry Pi 4 Model B offered the required architectural integrity to test the software hypothesis, thereby confirming that the local OpenCV stereo block matching could run reliably within the 100ms latency constraint.

Furthermore, housing the system within a "smart hat" rather than smart glasses distributed the payload across the cranium, mitigating the localized nose and ear fatigue common in wearable displays. Placing the dual OV5640 cameras on the hat's brim allowed for a 13cm baseline separation. This arrangement effectively mimics the natural human interpupillary distance, which proved highly effective for generating human-scale depth mapping required for walking navigation.

C. Failure-Case Study

During development and experimentation these are the noticed failure-cases of the study:

1. **Low textured surfaces** - On low textured surfaces the depth map generated contains more holes resulting in a lack of audio feedback.

2. **Change of lighting** - When lighting changes, the camera sensors adjust exposures at slightly different rates causing a noticeable noise in the generated depth map due to the temporal mismatch.

D. Implications, Impact, and Future Work

This research proves the viability of the concept of transforming visual data into controllable audio form, and its applications are not limited to physical navigation but can be applied in digital accessibility on social media platforms and object recognition. The successful implementation of the API integration proves the extensibility of the system. Although the optimization of the hardware was not the main aim of this research, the main idea of selective filtering can be applied to the existing hardware platform of smart glasses with LiDAR technology. Future versions will include eye-tracking technology for hands-free operation and a combination of audio and tactile feedback to further reduce sensory fatigue and accommodate users with limited motor skills.

The limitations of this research are that it is dependent on head movement and gesture commands, which only provide a solution to the visual-to-audio translation problem. Future research will focus on the integration of LiDAR technology for accurate depth perception, other methods of operation such as voice commands, and long-term usability studies with visually impaired users.

VII. CONCLUSION

This project presents Stesis, a wearable AIoT navigation hat that aimed to address sensory overload in assistive systems by introducing selective filtering instead of continuous auditory streaming. With a Raspberry Pi 4 and a local Node.js/Ollama server, the two-mode system demonstrated sub-100ms latency for binaural depth mapping. The quantitative benchmarking showed that Stesis enhanced navigation speed by 31.2% and decreased obstacle collisions by 65.3% compared to continuous streaming. Most importantly, although 3D sonification raised a specific mental demand, the removal of repetitive physical scanning decreased overall cognitive workload by 41.8%. Even with difficulties in low-texture environments, the findings confirm that physical control over sensory input hastens subconscious integration. The modular framework paves the way for future multimodal sensor fusion, advancing assistive technology toward SDG 10.

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